

Yibo Yuan

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EDUCATION

Xi'an Jiaotong University

Xi'an, China

B.Eng. in Biomedical Engineering, School of Life Science and Technology

2021 - 2025

- **Relevant courses:** Pattern Recognition & Machine Learning (85/100), Biomedical Sensors & Instruments (84/100), Medical Device R&D Management & Certification (100/100), Data Structures & Algorithms (80/100)

RESEARCH EXPERIENCE

Systems Social Neuroscience Lab, Westlake University

Aug 2025 - Present

Research Assistant to Professor Ding Liu (Full-time)

Programmable Biomimetic Robotic Mouse for Social Interaction Mar 2026 - Present

- Conceived and lead the project and initiated its cross-institutional collaboration with Prof. Qing Shi's biomimetic robotics team. Designed cue-decomposition and reunion-probe paradigms, built the behavioral arena, and personally conducted experiments to test which sensory and response cues support social interaction.
- Built a sound-attenuated platform with synchronized multi-camera and ultrasonic recording. Developed perception, planning, and host control, including real-time semantic segmentation, skeletal tracking, behavior classification, ROI-constrained visual redirection, and serial communication.
- Coordinated integration with collaborators' low-level controller and PPO-based locomotion. Designed manipulations of movement, touch, temperature, odor, and responsiveness, including controlled contact timing, duration, and tactile properties.

MORI: Relational Power in Human-AI Companionship Aug 2026 - Present

- Conceived and led a 105-participant experiment on how people grant, revise, and withdraw distinct powers from an AI companion, with Sichao Wang and Prof. Ding Liu. First-author manuscript under review at ACM CHI 2027.
- Designed a fixed 20-year fictional family narrative spanning six powers, four governance arrangements, and three judgment stages; collected choices, within-trial revisions, decision times, and open-ended explanations.
- Analyzed power-specific and bidirectional boundary revision across relationship contexts; developed follow-up work on long-term AI companionship and the distinction between trust, successful assistance, and permission to act.

Neural Population Dynamics of Social Need Nov 2025 - Mar 2026

- Analyzed miniscope calcium-imaging and naturalistic behavioral data to study how isolation and reunion reshape medial preoptic nucleus (MPN) activity. Independently proposed and implemented a baseline-referenced population-state framework for cross-session comparisons without registering individual neurons across days.
- Combined population geometry (CKA and subspace alignment), behavioral hazard modeling, and cross-validated neural decoding to assess social-need-related changes in population sensitivity, with motion controls and circular-shift/permutation tests.
- Developed an open-source semantic-segmentation workflow for quantitative social-contact analysis when low-resolution video and identity swaps limited pose estimation; integrated perspective correction, temporal alignment, tracking, and frame-level export.

Experimental Systems for Motion-Sickness Research Aug 2025 - Oct 2025

- Designed and built an eight-channel STM32 behavioral-monitoring system integrating capacitive lickometry, LED control, temperature monitoring, real-time visualization, and serial data logging.
- Designed and assembled a programmable three-axis motion platform with stepper motors, configurable sinusoidal profiles, and synchronized recording for motion-sickness experiments.
- Modeled and 3D printed experimental apparatus in Blender, including custom components and adaptations of existing laboratory devices.

AI for Scientific Simulation and Discovery Lab, Prof. Tailin Wu's Group Mar 2026 - Present

- Work with Prof. Tailin Wu and Tengfei Xu on neural-signal reconstruction and turbulence-based modeling of cortical dynamics using resting-state fMRI; develop models of cortical phase singularities.
- Analyze phase gradients and vortex trajectories using BrainVortexToolbox from Prof. Pulin Gong's group at the University of Sydney, supporting comparison of simulated and observed cortical dynamics.

Illinois Institute of Technology, Remote Collaboration**Nov 2025 - Present***Research Collaborator, Supervisor: Professor Ren Wang****Cross-Subject EEG-fNIRS Decoding***

- Developed a heterogeneous-expert EEG-fNIRS fusion framework for cross-subject brain-state decoding under participant-disjoint and few-shot evaluation protocols; investigated probability calibration and generalization to new participants.
- Analyzed modality reliability and failure cases in multimodal fusion to inform reliability-aware gating and fallback to individual modalities.

ReGround: Experience Reuse under Changing Authority Jul 2026 - Present

- First-author research with Sichao Wang, Yang Su, and Prof. Ren Wang on reusing collaboration experience when an AI agent's role or authority changes. Manuscript under review at ICLR 2027.
- The framework links unmet requirements, their blockers, and completion constraints to memory retrieval and authorized planning, with synthetic-benchmark evaluations of planning and language-based authority updates.

Institute of Health and Rehabilitation Sciences, Xi'an Jiaotong University**2025***Graduation Project, Supervisors: Professors Xiang Chen and Jin Li****Portable Bioimpedance Spectroscopy***

- Designed and bench-tested a portable multi-frequency bioimpedance prototype for tissue-water-content sensing using the Cole-Cole model, as a bachelor's graduation project.
- Integrated an STM32 microcontroller and AD5933 impedance converter; completed literature review, system architecture, PCB design, hardware debugging, and bench measurements.

Institute of Health and Rehabilitation Sciences, Xi'an Jiaotong University**2024***Project Leader, Supervisor: Professor Jin Li****Eight-Channel EEG Acquisition System***

- Led architecture and PCB design for an ADS1299/STM32F103C8T6 EEG acquisition platform, selecting components and system architecture through literature review.
- Designed power/reference circuits, electrode input protection, anti-aliasing networks, bias drive, and test-signal paths.
- Completed board bring-up and basic power, reference, and clock checks; specified further SPI/DRDY, test-tone, noise, and CMRR validation.

Institute of Mitochondrial Biomedicine, Xi'an Jiaotong University**Jan 2022 - Oct 2022***Lab Member, Supervisor: Professor Dan Tan****XJTU iGEM Synthetic Biology Project***

- Contributed plasmid design, PCR/RT-PCR, nucleic-acid purification, Western blotting, electrophoresis, and literature analysis to a synthetic E. coli project; the XJTU team received an iGEM Gold Medal.

INTERNSHIP EXPERIENCE**Zhen Tec Technology Inc.****Jun 2024 - Jul 2024***EEG Data Analyst*

- Designed IIR/FIR EEG filters and time/frequency features using MNE, including Higuchi fractal dimension and Hjorth parameters; studied PCA/ICA artifact handling.
- Supported a PyTorch CNN-LSTM driver-fatigue model and assembled open EEG datasets for analysis and visualization.

Collegeland Education Co. Ltd.**May 2024 - Sept 2024***AI Assistant Developer*

- Built data-collection and processing tools and supported educational LLM workflows, including prompting, hallucination handling, and adversarial testing.

First Affiliated Hospital (FAH) of Xi'an Jiaotong University (XJTU)**May 2023 - Jun 2023***Clinical Laboratory Intern*

- Observed rehabilitation and imaging workflows, assessed usability constraints in a hand-nerve-injury rehabilitation device, and proposed improvements to patient wearability.

EXTRACURRICULAR ACTIVITIES**Medical and Engineering Innovators****Jan 2024 - Present***Council Member, External Relations*

- Led alumni liaison and supported a Technology and Finance forum at Xi'an Jiaotong University; proposed a hospital-university translational project on olfactory dysfunction.

PUBLICATIONS AND MANUSCRIPTS

Yuan, Y., Wang, S., & Liu, D. (2026). MORI: Relational Power in Long-Term Human-AI Companionship. First-author manuscript under review at ACM CHI 2027.

Yuan, Y., Wang, S., Su, Y., & Wang, R. (2026). Re-Grounding Relational Experience under Changing Authority. First-author manuscript under review at ICLR 2027.

Dai, Y. Y., Wei, C., **Yuan, Y.**, Rahman, M. M., & Liu, D. (2026). In vivo calcium imaging with a miniaturized microscope in the hypothalamus for understanding social behaviors in mice. Journal of Visualized Experiments, (229), e70401. [doi:10.3791/70401](https://doi.org/10.3791/70401)

HONORS AND AWARDS

- National College Students' Innovation and Entrepreneurship Training Program: projects on OTU deubiquitinating enzymes (2024) and the human brain single-cell atlas (2023).
- Second Prize, 35th Tengfei Cup Innovation Track (2024); Gold Prize, 34th Tengfei Cup Entrepreneurship Track (2023).
- Second Prize, Xi'an Jiaotong University Mathematical Modeling Competition (2023); iGEM Gold Medal, XJTU team (2022).

OPEN SOURCE TOOLS

- [Mouse-trajectory-tracking](#) - Semantic-segmentation and tracking workflow for irregular arenas, contact quantification, perspective correction, and temporal synchronization.
- [NeuronTracker](#) - Human-in-the-loop axon-growth tracing with biologically motivated path constraints, developed for a collaboration with Prof. Rui Fang's lab.
- [Lickometer](#) - Eight-channel STM32 capacitive sensing system with embedded firmware, serial communication, real-time visualization, and quantitative export.
- Circuit Visualizer: Web-based 3D viewer for mouse brain circuits with AI-assisted query, built with a doctoral student in Prof. Ding Liu's lab.
- Macaque behavioral analysis: Customized mask-based 2.5D workflow for cage calibration, trajectory proxies, activity fragmentation, and spatial occupancy, developed for a collaboration with Prof. Xiaodong Liu's team.

RESEARCH AND TECHNICAL SKILLS

- **Neural and behavioral analysis:** Miniscope calcium-imaging data, population geometry, cross-validated decoding, representational similarity, behavioral hazard analysis, permutation controls, and video-based contact quantification.
- **Human-AI interaction research:** Human-participant experiment design, interactive narrative tasks, analysis of choices, boundary revisions and decision times, and interpretation of open-ended responses.
- **Experimental systems and robotics:** Vision-based perception and tracking, host-side planning and control, STM32 firmware, PCB design and debugging, serial communication, synchronized multimodal recording, Blender, and 3D printing.
- **Multimodal decoding and evaluation:** EEG-fNIRS fusion, time-frequency features, participant-disjoint and few-shot evaluation, probability calibration, modality-reliability assessment, and failure analysis.
- **Programming and software:** Python, C, MATLAB, PyTorch, scikit-learn, MNE, NumPy, pandas, OpenCV, MMSegmentation, Git/GitHub, and Linux; React and TypeScript for interactive research tools.
- **Additional laboratory training:** Mouse stereotactic surgery, perfusion and cryosectioning; PCR/RT-PCR, nucleic-acid extraction, and Western blotting.